Baseline-First Audit of Shape-Based Objectives in Square-Lattice Tight-Binding Quantum Dots

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We propose and apply a baseline-first audit hierarchy for shape-based objectives in a controlled square-lattice tight-binding model of finite superelliptic quantum dots. The audit uses existing closed-spectrum calculations, surrogate-model comparisons, inverse-screening tests, continuum-reference checks, tight-binding-only finite-size scaling diagnostics, and an open-transport contact-width diagnostic. The model uses superelliptic boundaries, direct Kwant spectra, and the low-energy convention $E_{\mathrm{kin},k} = E_k + 4$. Across these outputs, attractive observables are repeatedly absorbed by simpler explanations: confinement scale, aspect ratio, symmetry, boundary realization, reference uncertainty, or lead-dot coupling. A small multilayer perceptron does not establish a robust advantage over a physics-informed Ridge model under structured validation. Spectral inverse-screening objectives based on gap ratios and doublet splitting reduce to isotropy, anisotropy, or symmetry baselines. A finite-difference continuum reference is useful for calibration but not adequate as a high-precision shape-sensitive reference in the tested protocol. Tight-binding-only scaling remains inconclusive, and the tested open-dot conductance is more sensitive to contact width than to superellipse shape. The contribution is methodological and protocol-specific: the paper documents where apparent shape, ML, continuum-residual, and transport signals become unreliable before they can justify a device-design rule.

I. INTRODUCTION

Shape control is an appealing route for studying and designing quantum-dot spectra. Quantum dots and quantum billiards provide a natural setting for this question because confinement, symmetry, and boundary geometry can all affect low-energy states [1–4]. Tight-binding billiards and lattice quantum-dot models make these effects computationally explicit while retaining finite-size and boundary-realization features that are absent from ideal continuum sketches [5, 6]. These ingredients make a square-lattice tight-binding model of superelliptic quantum dots a useful controlled benchmark: the Hamiltonian is simple, the geometry family is interpretable, and spectra and conductance can be computed with standard sparse-eigensolver and quantum-transport tools [7, 8]. At the same time, this simplicity creates a demanding test of any proposed shape objective. If an apparent signal is already explained by size, area, aspect ratio, symmetry, degeneracy, boundary pixelation, reference-solver uncertainty, or lead coupling, then it should not be presented as inverse design or as a new physical effect.

This article is a methodology-oriented limitations paper. It does not enlarge the parameter space, repair unsuccessful directions, or promote a new objective. Instead, it proposes and applies a baseline-first audit hierarchy to a fixed set of existing calculations in one finite superellipse tight-binding benchmark. The hierarchy asks whether an apparent shape or ML-assisted signal survives successively simpler explanations: confinement scale, area and perimeter, aspect ratio, isotropy and degeneracy, lattice boundary realization, reference-solver uncertainty, and contact geometry.

This kind of audit is needed because modern shape-controlled and ML-assisted workflows can make weak observables look successful. A surrogate may interpolate a dataset well while still learning size, aspect-ratio, or symmetry structure; inverse-design candidates may then inherit the same baseline rather than reveal a new physical effect. Classical surrogate-modeling practice therefore requires comparison against simple, structured alternatives, and spatially or hierarchically structured data require validation splits that match the intended use [9–13].

Shape-controlled quantum dots are especially vulnerable to this failure mode. Low-energy levels of confined systems are strongly governed by dimensional scaling; near the bottom of a square-lattice tight-binding band, the dispersion has the expected quadratic form. In addition, size, area, aspect ratio, symmetry, finite-lattice boundary realization, and transport contacts can all mimic or absorb shape dependence. A baseline-first hierarchy is therefore a necessary diagnostic before claiming a shape effect, a positive ML result, or a device-design rule.

The main message is conservative and protocol-specific. In the tested square-lattice tight-binding model of finite superelliptic quantum dots, several plausible shape, ML, continuum-reference, and transport objectives do not support robust positive claims after baseline checks. This does not imply that geometry-based quantum design, machine learning, finite-size analysis, or transport studies are useless in general. The useful output is the documented audit hierarchy and the associated failure modes in this controlled benchmark.

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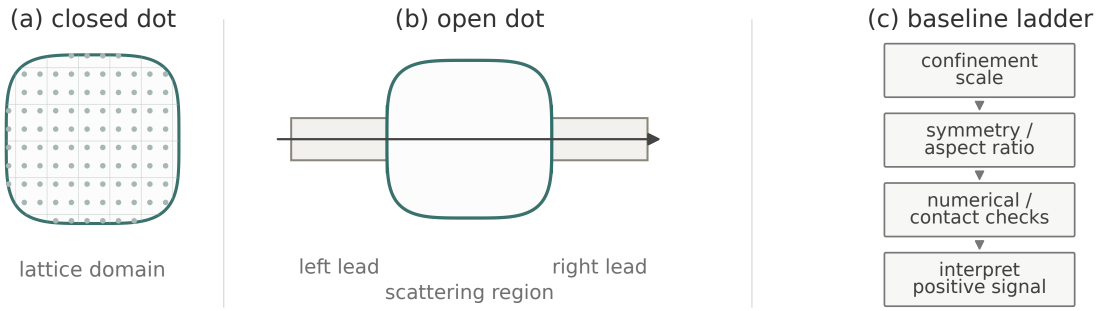

FIG. 1. Schematic of the closed finite-lattice superellipse dot, the corresponding open-dot transport geometry with separate left and right leads, and the baseline ladder used before interpreting shape-dependent observables. The ladder is conceptual: it describes the order of checks, not a new numerical calculation.

II. MODEL AND NUMERICAL PROTOCOL

The common model is a spinless nearest-neighbor tight-binding Hamiltonian on a square lattice,

$$H = \sum_i \epsilon_i c_i^\dagger c_i + \sum_{\langle i,j\rangle} t\left(c_i^\dagger c_j + c_j^\dagger c_i\right), \tag{1}$$

with onsite energy $\epsilon_i = 0$ and hopping $t = -1$. The bottom of the infinite square-lattice band is at $E = -4$, so low-energy spectral quantities are reported with the kinetic convention

$$E_{\mathrm{kin},k} = E_k + 4. \tag{2}$$

Closed-dot spectra are computed by direct sparse diagonalization of finalized Kwant Hamiltonians [7, 14, 15]. The same tight-binding convention underlies the transport diagnostic, where two-terminal spinless conductance is written as $G = T$ in Landauer units, with lead self-energies handled by Kwant [8]. For two conductance curves sampled on the same energy grid, the raw curve distance used in the transport diagnostic is the normalized RMS distance

$$R_F[X] = \left[N_f^{-1}\sum_{i\in F} X(E_i)^2\right]^{1/2}, \tag{3}$$

$$D_{\mathrm{raw}}(G_a, G_b) = \frac{R_F[G_a - G_b]}{\left(R_F[G_a]^2 + R_F[G_b]^2\right)^{1/2}}, \tag{4}$$

where F is the set of energy samples where both curves are finite and $N_f = |F|$. The contact-width effect is $D_{\mathrm{raw}}(G_{n=2,W=4}, G_{n=2,W=10})$. The shape effect at fixed W is the mean pairwise D_{raw} among the tested $n = \{1.2, 2.0, 4.0\}$ curves at that W. A separate shift/rescale diagnostic minimizes the same distance after integer index shifts up to 12 samples and an affine rescaling of one curve; those values are reported only as a diagnostic check. The plotted transport bars use the raw normalized distances in Eq. (4).

The geometry family is the lattice realization of a superellipse,

$$\left|\frac{x}{a}\right|^n + \left|\frac{y}{b}\right|^n \le 1, \qquad b = a r_{\mathrm{AR}}, \tag{5}$$

where a sets the major size, r_{AR} is the aspect ratio, and n controls the boundary shape. The completed closed-spectrum benchmark contains 140 geometries with $n = \{1.2, 2.0, 3.0, 4.0\}$, $a = \{24, 27, 30, 33, 36\}$, and $r_{\mathrm{AR}} \in \{0.67, 0.72, 0.78, 0.83, 0.89, 0.94, 1.0\}$. The primary spectral targets in the benchmark analysis are E_0 and $dE_1 = E_1 - E_0$; higher gaps are retained as diagnostics because degeneracies and level ordering make them less stable targets.

The analysis uses only pre-existing outputs listed in the source inventory. All numerical values used in the audit are traceable to versioned repository outputs there; the inventory separates confirmed numerical values and file provenance from the interpretation used in this manuscript. The manuscript does not introduce additional parameter scans beyond these outputs. The evidence is grouped by protocol:

1. a closed-spectrum benchmark dataset and Ridge versus small-MLP comparison;
2. one-shot Q inverse screening and preregistered S-objective screening;
3. a direct-Kwant magnetic-response ranking check;
4. a strict continuum-residual protocol, including a circular Bessel anchor and finite-difference reference checks;
5. a tight-binding-only self-scaling diagnostic after the FD reference became insufficient for shape-sensitive residuals;
6. an open-transport contact-width diagnostic.

The shared analysis rule is baseline-first. Before interpreting an observable as shape physics or inverse-design evidence, the observable must be compared against the simplest competing explanations: confinement scale, area, perimeter, aspect ratio, isotropy, symmetry and degeneracy, lattice boundary realization, reference-solver uncertainty, and, for transport, lead-dot coupling. Machine learning is treated as a candidate generator or surrogate approximation tool, not as physical truth.

III. RESULTS AND DISCUSSION

A. Closed-spectrum surrogate modeling

The closed-spectrum benchmark establishes that the direct tight-binding workflow is physically calibrated at the level needed for a conservative surrogate study, but it does not establish an ML novelty claim. Two scale checks play different roles. The circular Bessel-scale check tests the low-energy convention and circular calibration; it gives an error of about 2.03%. The fixed-shape scaling diagnostic tests whether the expected leading a^{-2} confinement scaling is consistent across the five tested sizes; it finds a maximum deviation of about 2.12% for $(E_0 + 4)a^2$. These are different diagnostics and should not be interpreted as independent measurements of the same error. Together they support the benchmark convention and leading confinement scale, not a nontrivial shape-design rule.

Within this dataset, the physics-informed Ridge model remains the preferred surrogate. A small multilayer perceptron with physics-motivated features passes the cell-level success rule in 10 of 16 main comparison cells, but only in 3 of 8 leave-one-aspect-out cells and 7 of 8 leave-one-aspect-ratio-out cells. Thus the preregistered robust-success criterion is not met. The numerical scale of the Ridge error is already small: the relative MAE is 2.01–2.60% of the characteristic E_{kin} for E_0 and 0.78–1.15% of the characteristic dE_1 gap. The MLP has individual winning cells, including a large favorable improvement for one E_0 protocol, but it also has strongly negative cells for dE_1. The safe conclusion is that the MLP does not robustly outperform a physically motivated linear baseline in the structured validation used here.

This result matters for later inverse-screening attempts. If the surrogate is not a robust model result, then surrogate-generated candidates must be treated only as proposals. Direct Kwant verification and comparison with physics baselines remain the source of any claim.

B. Spectral objectives and inverse-screening limitations

The first inverse-screening direction used the ratio $Q = dE_1/(E_0 + 4)$ as a one-shot surrogate-guided objective inside the verified training-domain ranges. The report explicitly treats Ridge predictions as candidate generators only, with final candidate and baseline values coming from direct Kwant calculations or already computed training rows. The outcome is negative for each fixed n class. The symmetry-optimum follow-up further shows that the earlier top inverse-screening candidate and the isotropic same-n baseline are the same discrete geometry for every tested n.

The preregistered S-objective line tests $S = (E_2 - E_1)/(E_0 + 4)$ under a frozen rule set. Its own report emphasizes that observed doublet splitting is not a discovery: the question is whether surrogate-guided screening beats strong baselines. It does not. At the primary setting $\alpha = 0.95$, 0 of 4 fixed n values pass; at the secondary setting $\alpha = 0.90$, the result also remains 0 of 4. The strongest baseline is the simple anisotropy heuristic for the reported rows.

Together these results limit a broad class of spectral ratio objectives. Gap ratios and doublet-splitting measures can encode obvious geometric trends such as isotropy, aspect ratio, and symmetry lifting. Without a stronger baseline ladder, they should not be used as evidence for inverse design or new shape-dependent physics.

C. Magnetic-response ranking limitation

The magnetic-response ranking analysis was a direct-Kwant check, not an ML task and not an extension of the Q/S objectives. It tested four representative geometries at sizes $a = \{30, 36\}$, with weak-field flux parameters $\alpha = (0.0, 0.00125, 0.0025, 0.005)$ and diagnostic values $\alpha = (0.01, 0.02, 0.04)$. Numerical checks passed: the $\alpha = 0$ spectra were reproduced, gauge invariance was verified with maximum reported difference $1.865174681370263 \times 10^{-14}$, Hermiticity and finite sorted spectra checks passed, and eigenvalue imaginary parts were controlled.

The proposed geometry-ranking signal is not supported. The analysis found 0 thresholded pairwise ranking crossovers and 0 crossovers surviving the size-stability filters. Three robustness-divergence candidates were identified, but 0 survived those filters. The apparent candidates are attributable to circle or ellipse symmetry-lifting baselines. The field-dependent calculation is numerically controlled, but this tested protocol does not produce a non-baseline geometry-ranking signal.

D. Continuum-reference limitation

The continuum-residual analysis asked whether tight-binding spectra contain a nontrivial superellipse-exponent-dependent finite-lattice convergence residual after subtracting ordinary continuum, Weyl, perimeter, pixelation, and discretization baselines. The protocol deliberately forbids weak claims based on the leading $1/a^2$

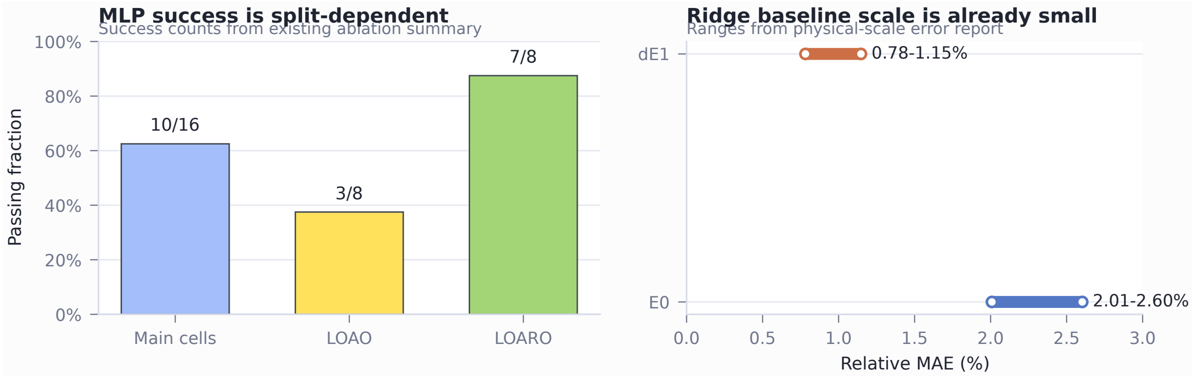

FIG. 2. Surrogate-model diagnostic from existing closed-spectrum outputs. The left panel summarizes the MLP success counts under progressively more structured validation splits. The right panel summarizes the Ridge relative-MAE ranges used as the conservative baseline scale. The figure is a visualization of repository outputs only; it does not introduce new training runs.

scale, generic boundary effects, ML accuracy, or largest-size tight-binding extrapolation. Its first check is a circular Bessel anchor for $n = 2.0$, $r_{\mathrm{AR}} = 1.0$, with sizes $a = \{24, 30, 36, 48, 60, 72, 96\}$ and levels E_0 through E_5.

That anchor passes its stability review, which is useful but narrow. It calibrates the low-energy circular case; it does not establish a shape-sensitive continuum residual for $n \neq 2.0$. The finite-difference reference step then validates an embedded-mask Dirichlet Laplacian on grids up to $N_{\mathrm{grid}} = 251$, where the selected circular ground-state relative error versus the Bessel value is 0.0052206740942014095. This is adequate as an initial reference check, but the later $N_{\mathrm{grid}} = 501$ convergence check changes the conclusion for shape-sensitive use.

The N501 check uses the near-geometric refinement triple $(126, 251, 501)$, corresponding approximately to $h = \{0.016, 0.008, 0.004\}$. Although the raw circle ground-state relative error versus Bessel improves to 0.0026734921480308575, the non-circular reference uncertainties remain too large for the downstream residual claim: 0.004721625115920378 5 for the $n = 1.2$ level-zero reference and 0.0021923716263753637 for the $n = 4.0$ level-zero reference. Both exceed the 0.001 reference-uncertainty criterion. The safe conclusion is not that FD is generally invalid, but that this embedded-mask FD reference should not be treated as continuum truth for the intended shape-sensitive residual.

E. TB-only finite-size scaling limitation

After the FD reference was found insufficient for the continuum-residual analysis, the TB-only self-scaling diagnostic used direct tight-binding ground-state values and geometric diagnostics only. No FD reference values were loaded, no TB-vs-FD residuals were computed, and no ML, Q, or S objectives were used. The domain was $r_{\mathrm{AR}} = 1.0$, $n = (1.2, 2.0, 4.0)$, and $a = (24.0, 30.0, 36.0, 48.0, 60.0, 72.0, 96.0)$.

The circular calibration is good: for $n = 2.0$, the effective-radius model gives a relative error of 7.21×10^{-4} against $j_{01}^2 = 5.783185962946783$. Fitted effective-radius parameters are also reported for the three n values. The limiting coefficients are approximately 8.03, 5.779, and 5.049, with offsets approximately 0.279, 0.277, and 0.130 for $n = 1.2, 2.0, 4.0$, respectively. These values show ordinary shape dependence of continuum-like eigenvalues and finite-size offsets. They are not, by themselves, a supported shape-residual signal.

The potential residual exponent is unstable. The reported power-law values are approximately $p_{\mathrm{TB}} = 0.10, 0.53, 1.05$ for $n = 1.2, 2.0, 4.0$, but the power-law leave-one-size-out stability test is not accepted for any of the three n values. The corresponding leave-one-size-out p ranges are about 0.67, 1.73, and 1.01, respectively, and the $n = 1.2$ fit sits at the lower bound of the implemented power-law fit. The existing diagnostic therefore reports p_{TB} as an instability indicator, not as a physical exponent with a defensible uncertainty interval. Simple geometric baselines do not cleanly explain a robust residual, but the residual itself is not stable enough to support a shape-residual claim. The supported scientific result is inconclusive.

F. Open-transport limitation

The open-transport diagnostic changes the observable from closed spectra to two-terminal conductance. This is physically richer, but it introduces a new baseline: contact geometry. The tested setup uses square-lattice strip leads, the spinless convention $G = T$, a superellipse dot with $a = 30.0$ and $r_{\mathrm{AR}} = 1.0$, lead widths $W = 4$ and $W = 10$, and an energy window $[-3.8, -3.0]$ sampled at 241 points. The curve distance is the raw normalized RMS distance defined in Eq. (4); all raw distances are computed on the same energy grid.

The central result is that contact width dominates

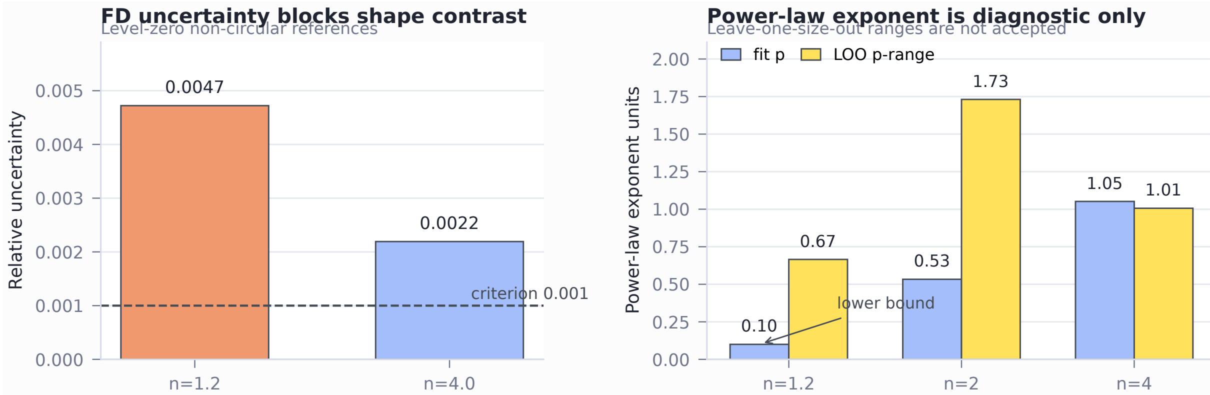

FIG. 3. Continuum-reference and TB-only scaling diagnostics from existing outputs. The FD reference uncertainties for non-circular shapes exceed the 10^{-3} criterion used before shape-sensitive residual analysis. The TB-only power-law exponents are accompanied by large leave-one-size-out ranges and are not treated as physical exponents.

the tested shape effect. For $n = 2.0$, the raw distance between $W = 4$ and $W = 10$ conductance curves is 0.6589809313045025. The mean pairwise shape distances are 0.5810925658201803 at $W = 4$ and 0.5122909816541278 at $W = 10$. The shift/rescale diagnostic gives 0.24323761388169962 for the contact comparison and 0.49646370215309615 and 0.2802443376667481 for the two shape comparisons. Mode thresholds do not explain the features, the straight-channel baseline is computed but does not remove the result, and an energy-shift/rescale collapse does not remove it either. The limitation is therefore not a trivial numerical failure. It is a physical interpretation problem: in this open-dot geometry, lead-dot coupling is at least as important as the shape parameter being tested.

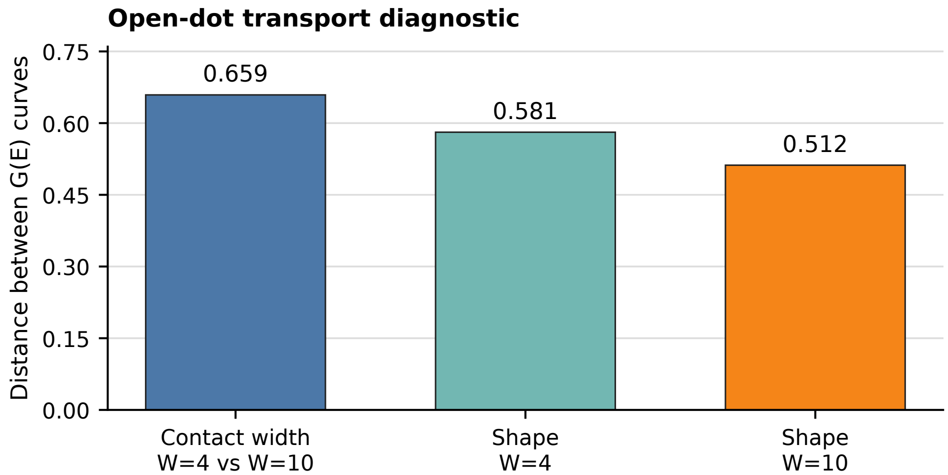

FIG. 4. Compact summary of the open-transport diagnostic. In the tested setup, the contact-width effect for $n = 2.0$ is larger than the raw normalized shape-effect summaries at both lead widths. The plotted values are rounded from the source inventory and use Eq. (4).

TABLE I. Merged summary of tested analyses, diagnostic values, limiting baselines, and safe conclusions. Exact values and provenance are listed in the source inventory; this table uses rounded values only to keep the audit readable.

Analysis	Observable / protocol	Key diagnostic value	Main baseline or limitation	Result	Safe conclusion
Closed-spectrum Ridge/MLP	E_0, dE_1 under structured validation	Ridge relative MAE 2.01–2.60% for E_0 and 0.78–1.15% for dE_1; MLP success 10/16, 3/8, 7/8	Physics-informed Ridge and split-aware validation	ML advantage not established	Ridge remains the preferred conservative surrogate; direct Kwant remains the reference.
Q inverse screening	$dE_1/(E_0+4)$	Success false for each fixed n; top candidate equals same-n isotropic baseline	Isotropy and same-n baselines	Not supported	Q candidates do not justify inverse-design claims.
S objective	$(E_2-E_1)/(E_0+4)$	0/4 pass at $\alpha = 0.95$ and 0/4 at $\alpha = 0.90$	Simple anisotropy and strongest feasible baselines	Not supported	S behaves as an anisotropy diagnostic in the tested domain.
Magnetic-response ranking	Weak-field level rankings	Thresholded crossovers 0; surviving crossovers 0; robustness divergences surviving filters 0	Symmetry lifting and zero-field/aspect-ratio ranking	Baseline limited	Field-dependent spectra are controlled numerically, but no non-baseline design signal is supported.
FD continuum reference	Embedded-mask FD reference for shape-sensitive residuals	Non-circular level-zero relative uncertainties 0.00472 and 0.00219 exceed 0.001	Reference-solver uncertainty	Inconclusive	Current FD reference should not be treated as continuum truth for shape-sensitive residuals.
TB-only scaling	Ground-state finite-size scaling	$n = 2$ calibration 7.21×10^{-4}; $p_{\mathrm{TB}} = 0.10, 0.53, 1.05$ with unstable LOO ranges	Unstable residual exponent and insufficient surviving signal	Inconclusive	The diagnostic does not support a continuum-residual claim.
Open transport diagnostic	$G(E)$ curves, $W = 4, 10$	Contact distance 0.659; shape distances 0.581 and 0.512	Contact-width dependence and lead-dot coupling	Contact dominated	Transport interpretation requires tighter lead-dot coupling control before a shape claim.

IV. GENERAL LESSONS

The common lesson is not that shape is irrelevant. The tested outputs must be read through simple baselines before they are treated as design evidence. Direct calculations remain useful for controlled numerical physics, but a shape-dependent observable is not automatically a shape-design rule. In this project, the strongest interpretation is often removed by a baseline that is simpler than the proposed objective.

The first layer is confinement scale. Because E_{kin} near the bottom of the band follows a continuum-like quadratic dispersion, many low levels and gaps inherit a leading size dependence. Any residual objective must show that it is not merely reporting this scale. The second layer is geometry compression: area, perimeter, aspect ratio, and site count can all absorb substantial variation before one reaches a shape-exponent-specific claim. The third layer is symmetry and degeneracy. Q, S, and magnetic ranking ideas all encounter this issue in different forms; isotropy and degeneracy lifting can explain the signal while still being too simple to count as inverse design.

The fourth layer is boundary realization. A smooth superellipse becomes a finite set of lattice sites, so boundary pixelation, area error, and sublattice imbalance can produce finite-size structure. These effects are not nuisances to hide; they are part of the model and must be included in the baseline ladder. The fifth layer is reference-solver uncertainty. The FD-reference sequence shows that passing a circular sanity check is not enough for a shape-sensitive continuum residual. A reference can be useful for calibration and still be too uncertain for the intended residual analysis.

Transport adds one more layer: contacts. Conductance is not only a property of the isolated dot shape. It depends on the coupling between the finite dot and the leads, on lead width, and on mode structure. The contact-width diagnostic therefore limits the shape-transport interpretation until the lead-dot coupling baseline is controlled.

These lessons also clarify the role of ML. A surrogate can be valuable for compression, interpolation, and candidate generation, especially when direct Kwant calculations are more expensive than model evaluation. But in the present dataset, a flexible model is not automatically more scientific than a Ridge baseline tied to confinement descriptors. ML should enter after the physical question survives baseline subtraction, not as a substitute for that survival.

The resulting baseline hierarchy is:

1. check size, area, perimeter, and aspect-ratio scaling;
2. check isotropy, symmetry, and degeneracy;
3. check lattice boundary realization, pixelation, and effective radius;
4. check reference-solver uncertainty before using residuals;
5. check contact geometry before interpreting transport;
6. only then interpret ML or inverse-screening outputs.

Applied to the existing outputs, this hierarchy supports a reproducible limitations benchmark rather than a new device-design rule.

V. CONCLUSION

This paper summarizes a baseline-first limitations analysis for finite superelliptic quantum dots on a square tight-binding lattice. The system remains useful as a controlled benchmark: the Hamiltonian convention is clear, the geometry family is compact, and the direct Kwant calculations provide a consistent reference for the tested finite systems. The audit shows, however, that several attractive objectives fail to support robust positive claims once simple baselines and methodological checks are applied.

For the closed spectra, the physics-informed Ridge model remains the preferred conservative surrogate, while a small MLP does not meet the robust structured validation criterion. Q and S inverse-screening objectives do not beat their isotropy, symmetry, or anisotropy baselines. The magnetic-response ranking check is explained by symmetry-lifting baselines. The embedded-mask finite-difference reference is not adequate as a high-precision continuum truth for shape-sensitive residuals, and the TB-only self-scaling fallback remains inconclusive. In the open-transport diagnostic, contact width dominates the tested shape comparison.

These results do not prove a universal no-go theorem for geometry-based quantum design, machine learning, continuum references, or transport objectives. They show that this particular superellipse metric search, with these protocols and outputs, does not justify a device-design rule. The main methodological result is therefore the audit hierarchy itself: apparent shape, ML, continuum-residual, and transport signals are interpreted only after confinement, geometry, symmetry, boundary realization, reference uncertainty, and contact baselines have been checked.

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